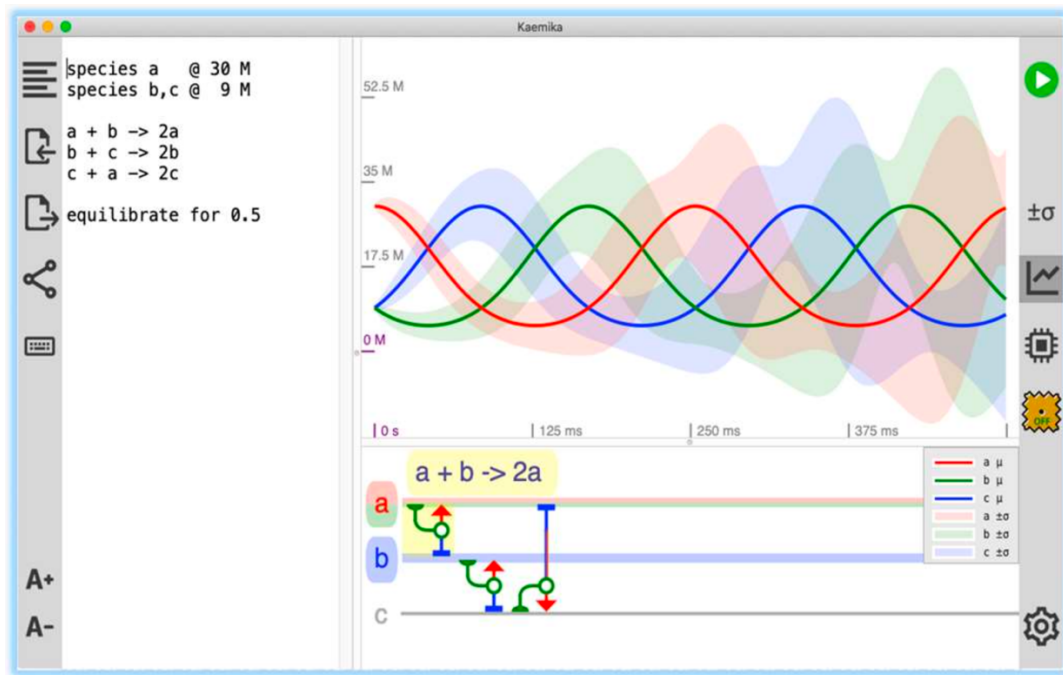




Biocomputing – part 2

AN INTRODUCTION TO KAEMIKA

By Professor Jan Madsen, Technical University of Denmark (DTU), November 2026.

Kaemika developed and distributed by Professor Luca Cardelli, University of Oxford, Oxford, UK

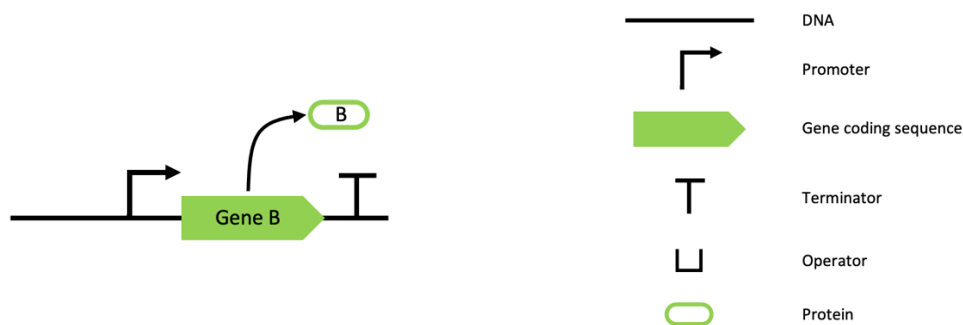
Experiments adopted for 27020 interdisciplinary bioengineering from a tutorial by Professor Geoff Baldwin, Imperial College, London, UK

4. Gene Regulation

In section 3 we saw how gene expression can be modeled in a very simple way.

Gene_B -> Protein_B

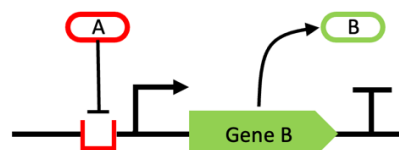
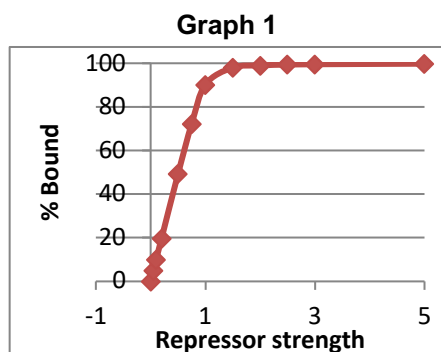
where the mRNA was abstracted away. We saw that the simulation results with and without mRNA were almost identical. This was an example of what is called a **constitutive expression**, i.e., the gene is continuously expressed. Using SBOL 3's graphical notation, this can be illustrated as below,



Very few genes are known to have a purely constitutive expression, most genes have their expression controlled by some outside signals (DNA-binding proteins, Temperature, metabolites, RNA molecules ...). In this section, we will particularly focus on the study of DNA-binding proteins, called **transcription factors**. These proteins, when binding to a promoter region, can either have an **activation effect** on the gene (positive control), or a **repression effect** (negative control). In prokaryotes, control of transcriptional initiation is the major point of regulation. In this part, we investigate one of the most common models used to describe this type of interactions, but first we will consider the molecular basis of gene repression.

4.1 Repressed gene expression

The object of a repressor is to give a switch-like characteristic to a promoter, so that it can be turned on and off very precisely. This is done by **transcription factors**; these are specific proteins that can bind to the DNA upstream from the promoter. The binding of a single protein to a DNA site follows a rectangular hyperbola. The 'sharpness' of the switch is given by a curve, graph 1 depicts a tight binding situation. The transcription factor A repressing the expression of protein B is shown below.

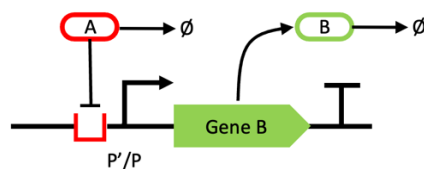


Before we can simulate the behavior of the repressor in Kaemika, we need to understand the mechanisms of the transcription factor and the operator. A transcription factor can bind to the operator site of the DNA, for repressing transcription factor, the consequence is that the RNAP cannot bind to the promoter and hence not express the gene. For the example in the figure, no A means that the RNAP can bind, and we will see an expression of B. Whereas the presents of A (in the right concentration) will prohibit the binding of RNAP and hence, no expression of B. This is in fact the logic behavior of an inverter gate (**INV**). Looking at what happens, we can see that the promoter P can either be **active** (we call this P') or **inactive** (we call this P).

Within a single bacterium, the specific promoter will either be active or inactive. However, when we use ODE's, we assume that we have much more bacteria and much more molecules. In this case some bacteria will have the promoter active while others are inactive, hence, P' and P can be viewed as a probability distribution, such that $P' + P$ is a constant which expresses the total number of such operators.

Ex4-1: Repressed gene expression (INV)

The following figure shows all the reactions needed to model the inverter in Kaemika.



Create the needed reactions in Kaemika. Assume that all reaction rates are 1 and the degradation rate of B is 0.1, while that of A is much slower, i.e. 0.0001.

1. Assume that there is no A initially and that the promoter hence is active, i.e., P' @ 1M. This simulation should give you the same behaviour as for the constitutive expression in Ex3-1.
2. We would now like to turn A 'on' after 50s. This can be done in Kaemika using the following trigger-command place before the equilibrate- command,

```
trigger A @ 1 mM when time > 50
```

3. Now simulate for 200s. Does your inverter behave as expected?
4. If not, try to change the rates and see if this makes any changes?
5. You can also try to toggle A several time (if you increase the concentration of A to 4 M, the effect will be more pronounced, i.e.,

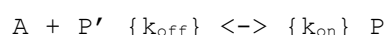
```
trigger A @ 4 M when time > 50
```

```
trigger A @ 0 M when time > 150
```

```
trigger A @ 4 M when time > 300
```

```
trigger A @ 0 M when time > 400
```

Let's take a closer look at what happens when the transcription factor A binds to the operator site of the DNA for the promoter. As before we will use the notation P' for the active promoter, i.e., the condition where transcription can take place to produce the protein B. When A binds to P', the promoter becomes inactive. This binding AP' we will denote P. We can write the reactions as,



The operator of the promoter can thus be either free P' or bound P , resulting in a conservation equation

$$P' + P = P_T$$

The rate of producing P is

$$dP/dt = k_{on} \cdot A \cdot P' - k_{off} \cdot P$$

which at steady state becomes

$$k_{on} \cdot A \cdot P' = k_{off} \cdot P \Leftrightarrow A \cdot P' = k_{off}/k_{on} \cdot P = k_{off}/k_{on} (P_T - P')$$

We call $k_{off}/k_{on} = k_d$ the dissociation constant, which describes how strong the binding of the transcription factors is (a high k_d means a weak binding). We get,

$$\begin{aligned} A \cdot P' &= k_d (P_T - P') \Leftrightarrow A \cdot P' = k_d \cdot P_T - k_d \cdot P' \Leftrightarrow A = k_d \cdot P_T / P' - k_d \\ &\Leftrightarrow \underline{P' / P_T = 1 / (A / k_d + 1)} \end{aligned}$$

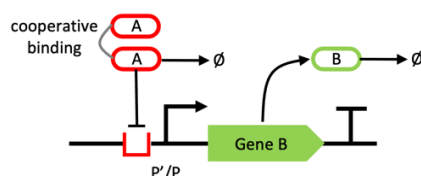
When $A=0$ the site is always free, i.e., $P' / P_T = 1$. The site has 50% chance of being free $P' / P_T = 1/2$, when $A=k_d$. When the operator site is free, RNAP can bind and transcribe with **maximum transcription rate** β . We now define the promoter activity as the rate of production times the probability that P' is free,

$$\text{promoter activity} = \beta / (A / k_d + 1)$$

So, for efficient repression enough repressor is needed so that the operator is almost always occupied with repressor. This happens when the repressor concentration greatly exceeds the dissociation constant, i.e., $A/k_d \gg 1$. Using the promoter activity, we can formulate the rate of production of B as a function of the bound transcription factor P' , i.e.,

$$B = f(A) = \beta / (A / k_d + 1)$$

A protein that binds very tightly will have a sharp curve, but it will also be difficult to reverse the binding, hence the switch characteristics may be poor. Nature overcomes this limitation by using multiple interactions. There is usually more than one operator site (positions where transcription factors bind), and usually multiple proteins that interact with each other as well as the operators. This gives rise to the phenomenon known as **cooperativity**. Cooperativity is very important in understanding gene regulation as it provides the switch like behaviour. The figure below shows how cooperativity can work, two A transcription factors bind and forms a stronger binding to the operator.



With n cooperating transcription factors ($n=2$ in the figure), we extend the rate function.

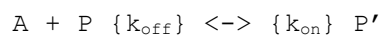
Ex4-2: Cooperatively repressed gene expression

Rewrite the reactions of the repressor in Ex4-1 using cooperativity of n A repressors.

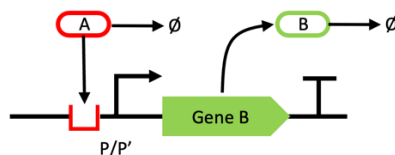
Do you see any change in the curves?

4.2 Activated gene expression

It is important to know that **activators** can perform the opposite function to repressors and can increase expression of a gene through productive interactions with the RNA polymerase. These can also be modelled by exchanging the active (P') and inactive (P) promoter. Which means that the promoter is inactive until the transcription factor A is present.



In the following figure, the transcription factor A activating the promoter is shown as an arrow from A to the operator site of the promoter.



Further, the promoter gets active (P') when the transcription factor binds to the operator site. In case of no transcription factor, B will be produced, but in considerably less concentration.

Activated gene expression can be used as a **sensor**; protein B is produced when a change in A occurs, or as an **enhancer**; protein B is produced in much higher concentration proportional to the concentration of A.

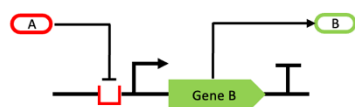
5. Modelling genetic logic circuits

In this section, we explore how repressed gene expression can be used to develop logic circuits. The following are a few examples,

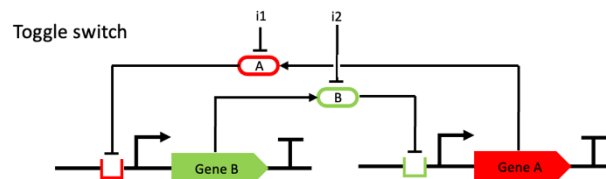
Ex5-1: Genetic logic gates using Hill function

Implement and simulate in Kaemika, the NOR and OR gates, and the Toggle switch. Do they behave as expected? Hint: for 2-input circuits it is easier to use P'/P model.

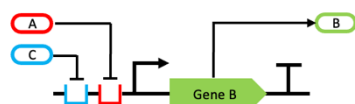
INV gate



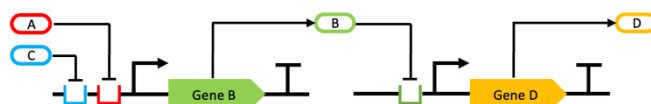
Toggle switch



NOR gate



OR gate



This exercise must be handed in through Learn. You will have to deliver your Kaemika code and plots to support that it works, as well as a short explanation of how you have approached the design – and in case you cannot get it to work, an explanation of what you think is the problem.